Ablation : ConvNeXt-Tiny — GELU | StochDepth | BatchNorm

Config: ImageNet- K pretrained | GELU activation | StochDepth | BatchNorm **Model:** ConvNe

Dataset: Rice Leaf Disease Identification Dataset

. Colab Setup (jalankan sekali per session)

In [1]:
```
from google.colab import drive
drive.mount('/content/drive')
import os
DRIVE_DIR = '/content/drive/MyDrive/rice-convnext/02_GELU_StochDepth_Batc
os.makedirs(DRIVE_DIR, exist_ok=True)
print(f'Drive folder ready: {DRIVE_DIR}')
```

```
Mounted at /content/drive
Drive folder ready: /content/drive/MyDrive/rice-convnext/
02_GELU_StochDepth_BatchNorm
```

In [2]:
```
from google.colab import files
import os, shutil
print('Upload kaggle.json...')
files.upload()
os.makedirs('/root/.kaggle', exist_ok=True)
shutil.move('kaggle.json', '/root/.kaggle/kaggle.json')
os.chmod('/root/.kaggle/kaggle.json', 0o600)
print('Kaggle API ready!')
```

```
Upload kaggle.json...
```

Upload widget is only available when the cell has been executed in the current browser session. Pl rerun this cell to enable.

```
Saving kaggle.json to kaggle.json
Kaggle API ready!
```

In [3]:
```
DATASET_DIR = '/content/drive/MyDrive/rice-convnext/dataset'
if os.path.exists(DATASET_DIR) and len(os.listdir(DATASET_DIR)) > 0:
    print('Dataset sudah ada, skip download.')
else:
    !pip install kaggle -q
    !kaggle datasets download -d wangxiaoqii/rice-leaf-disease-identifica
    !unzip -q /content/tmp/*.zip -d {DATASET_DIR}
    !rm -rf /content/tmp
    print('Dataset ready!')

import shutil
LOCAL_DATASET = '/content/dataset'
if not os.path.exists(LOCAL_DATASET):
    print('Copying dataset ke local SSD...')
    shutil.copytree(DATASET_DIR, LOCAL_DATASET)
    print('Done!')
else:
```

```
    print('Sudah ada di local.')
DATA_DIR = LOCAL_DATASET
```

```
Dataset sudah ada, skip download.
Copying dataset ke local SSD...
Done!
```

. Setup & Imports

In [4]:
```
import os, random, shutil
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from collections import Counter
from sklearn.metrics import classification_report, confusion_matrix
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, random_split
from torchvision import datasets, transforms, models

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
torch.cuda.manual_seed(SEED)
torch.backends.cudnn.deterministic = True

DEVICE = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Device: {DEVICE}')
if torch.cuda.is_available():
    print(f'GPU: {torch.cuda.get_device_name(0)}')
```

```
Device: cuda
GPU: Tesla T4
```

. Configuration

In [5]:
```
DATA_DIR      = '/content/dataset'
DRIVE_DIR     = '/content/drive/MyDrive/rice-convnext/02_GELU_StochDepth_
IMG_SIZE      = 224
BATCH_SIZE    = 64
NUM_EPOCHS    = 30
LR_INIT       = 1e-3
LR_FINAL      = 1e-6
WEIGHT_DECAY  = 0.05
NUM_WORKERS   = 2
IMAGENET_MEAN = [0.485, 0.456, 0.406]
IMAGENET_STD  = [0.229, 0.224, 0.225]
SAVE_PATH     = f'{DRIVE_DIR}/convnext_tiny_02_gelu_stochdepth_batchnorm.
EXP_NAME      = 'ConvNeXt-Tiny | GELU | StochDepth | BatchNorm'
print(f'Experiment: {EXP_NAME}')
```

```
Experiment: ConvNeXt-Tiny | GELU | StochDepth | BatchNorm
```

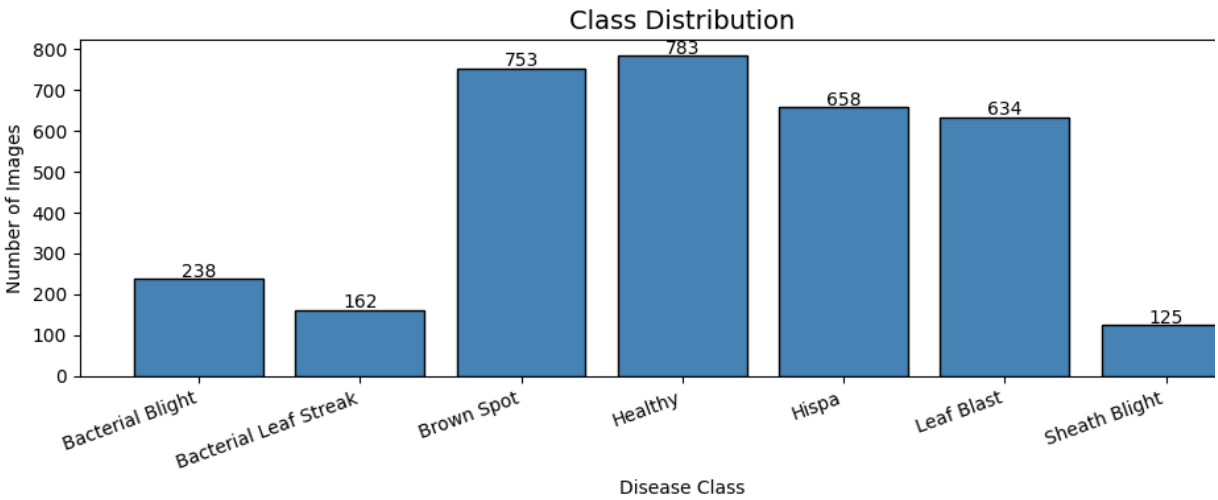
. Data Loading & Augmentation

In [6]:
```
train_transforms = transforms.Compose([
    transforms.RandomResizedCrop(IMG_SIZE),
    transforms.RandomHorizontalFlip(),
    transforms.RandomVerticalFlip(),
    transforms.RandomRotation(30),
    transforms.ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2),
    transforms.ToTensor(),
    transforms.Normalize(mean=IMAGENET_MEAN, std=IMAGENET_STD),
])
val_test_transforms = transforms.Compose([
    transforms.Resize(256),
    transforms.CenterCrop(IMG_SIZE),
    transforms.ToTensor(),
    transforms.Normalize(mean=IMAGENET_MEAN, std=IMAGENET_STD),
])

full_dataset = datasets.ImageFolder(DATA_DIR)
CLASS_NAMES  = full_dataset.classes
NUM_CLASSES  = len(CLASS_NAMES)
print(f'Classes ({NUM_CLASSES}): {CLASS_NAMES}')
print(f'Total images: {len(full_dataset)}')

total   = len(full_dataset)
n_train = int(0.6 * total)
n_val   = int(0.2 * total)
n_test  = total - n_train - n_val

generator = torch.Generator().manual_seed(SEED)
train_set, val_set, test_set = random_split(
    full_dataset, [n_train, n_val, n_test], generator=generator
)

train_set.dataset.transform = train_transforms
val_set.dataset.transform   = val_test_transforms
test_set.dataset.transform  = val_test_transforms

train_loader = DataLoader(train_set, batch_size=BATCH_SIZE, shuffle=True,
val_loader   = DataLoader(val_set,   batch_size=BATCH_SIZE, shuffle=False
test_loader  = DataLoader(test_set,  batch_size=BATCH_SIZE, shuffle=False
print(f'Train: {len(train_set)} | Val: {len(val_set)} | Test: {len(test_s
```

```
Classes (7): ['Bacterial Blight', 'Bacterial Leaf Streak', 'Brown Spot', 'Hea
'Hispa', 'Leaf Blast', 'Sheath Blight']
Total images: 3353
Train: 2011 | Val: 670 | Test: 672
```

In [7]:
```
labels_all = [full_dataset.targets[i] for i in range(len(full_dataset))]
counts = [Counter(labels_all)[i] for i in range(NUM_CLASSES)]
plt.figure(figsize=(10, 4))
bars = plt.bar(CLASS_NAMES, counts, color='steelblue', edgecolor='black')
plt.title('Class Distribution', fontsize=14)
plt.xlabel('Disease Class'); plt.ylabel('Number of Images')
plt.xticks(rotation=20, ha='right')
for bar, count in zip(bars, counts):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height()+5, str(cou
plt.tight_layout()
```

```
plt.savefig('class_distribution.png', dpi=150)
plt.show()
```



. Model — ConvNeXt-Tiny (GELU | StochDepth | BatchNorm)

In [8]:
```
class BatchNormNHWC(nn.Module):
    """BatchNorm2d wrapper that accepts NHWC input (used inside ConvNeXt
    def __init__(self, num_features):
        super().__init__()
        self.bn = nn.BatchNorm2d(num_features)
    def forward(self, x):  # x: (B, H, W, C)
        return self.bn(x.permute(0, 3, 1, 2)).permute(0, 2, 3, 1)

def replace_layernorm_with_batchnorm(module):
    """Recursively replace all LayerNorm variants with BatchNorm."""
    for name, child in module.named_children():
        if type(child).__name__ == 'LayerNorm2d':
            # Stem / downsampling: NCHW input → use BatchNorm2d
            C = child.normalized_shape[0]
            setattr(module, name, nn.BatchNorm2d(C))
        elif isinstance(child, nn.LayerNorm):
            # CNBlock (NHWC after Permute) → use BatchNormNHWC
            C = child.normalized_shape[0]
            setattr(module, name, BatchNormNHWC(C))
        else:
            replace_layernorm_with_batchnorm(child)

model = models.convnext_tiny(weights=models.ConvNeXt_Tiny_Weights.IMAGENE
in_features = model.classifier[2].in_features
model.classifier[2] = nn.Linear(in_features, NUM_CLASSES)

# Replace LayerNorm → BatchNorm
replace_layernorm_with_batchnorm(model)
# Fix classifier: after Flatten input is (B, C), use BatchNorm1d
if isinstance(model.classifier[2], BatchNormNHWC):
    C = model.classifier[2].bn.num_features
    model.classifier[2] = nn.BatchNorm1d(C)
print('Normalization: LayerNorm → BatchNorm')

model = model.to(DEVICE)
total_p     = sum(p.numel() for p in model.parameters())
```

```
trainable_p = sum(p.numel() for p in model.parameters() if p.requires_gra
print(f'Experiment : {EXP_NAME}')
print(f'Total params     : {total_p:,}')
print(f'Trainable params: {trainable_p:,}')
```

```
Downloading: "https://download.pytorch.org/models/convnext_tiny-983f1562.pth"
root/.cache/torch/hub/checkpoints/convnext_tiny-983f1562.pth
100%|██████████| 109M/109M [00:00<00:00, 182MB/s]
Normalization: LayerNorm → BatchNorm
Experiment : ConvNeXt-Tiny | GELU | StochDepth | BatchNorm
Total params     : 27,825,511
Trainable params: 27,825,511
```

. Training Setup

In [9]:

```
criterion = nn.CrossEntropyLoss()
optimizer = optim.AdamW(model.parameters(), lr=LR_INIT, weight_decay=WEIG

def get_scheduler(optimizer, num_epochs, lr_init, lr_final):
    warmup = optim.lr_scheduler.LinearLR(optimizer, start_factor=0.01, en
    cosine = optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=num_ep
    return optim.lr_scheduler.SequentialLR(optimizer, schedulers=[warmup,

scheduler = get_scheduler(optimizer, NUM_EPOCHS, LR_INIT, LR_FINAL)
```

. Training Loop

In [10]:

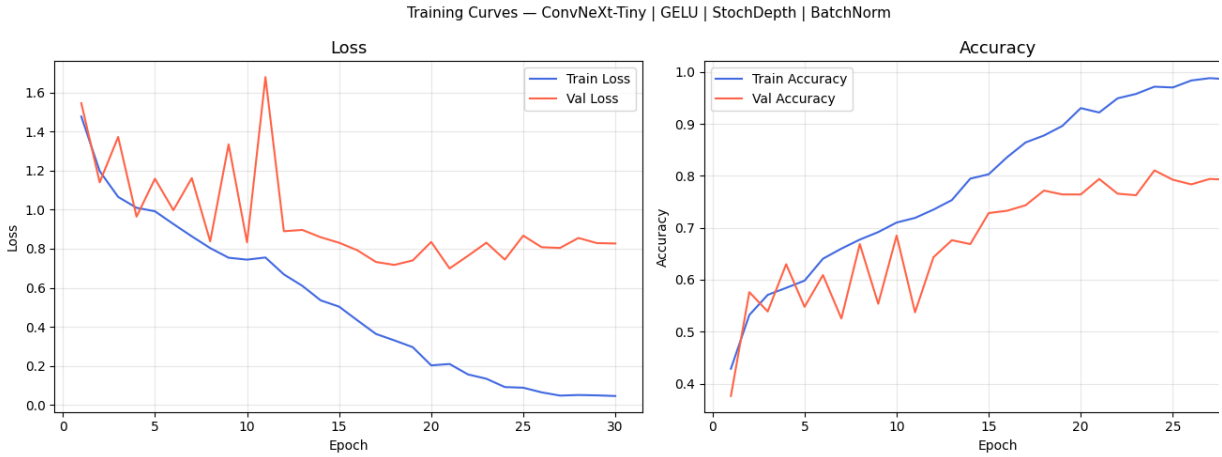
```
def train_one_epoch(model, loader, criterion, optimizer):
    model.train()
    running_loss, correct, total = 0.0, 0, 0
    for images, labels in loader:
        images, labels = images.to(DEVICE), labels.to(DEVICE)
        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()
        running_loss += loss.item() * images.size(0)
        _, predicted = outputs.max(1)
        correct += predicted.eq(labels).sum().item()
        total   += labels.size(0)
    return running_loss / total, correct / total

def evaluate(model, loader, criterion):
    model.eval()
    running_loss, correct, total = 0.0, 0, 0
    with torch.no_grad():
        for images, labels in loader:
            images, labels = images.to(DEVICE), labels.to(DEVICE)
            outputs = model(images)
            loss = criterion(outputs, labels)
            running_loss += loss.item() * images.size(0)
            _, predicted = outputs.max(1)
            correct += predicted.eq(labels).sum().item()
            total   += labels.size(0)
    return running_loss / total, correct / total
```

```
history = {'train_loss': [], 'train_acc': [], 'val_loss': [], 'val_acc':
best_val_acc = 0.0

for epoch in range(NUM_EPOCHS):
    train_loss, train_acc = train_one_epoch(model, train_loader, criterio
    val_loss,   val_acc   = evaluate(model, val_loader, criterion)
    scheduler.step()
    history['train_loss'].append(train_loss)
    history['train_acc'].append(train_acc)
    history['val_loss'].append(val_loss)
    history['val_acc'].append(val_acc)
    if val_acc > best_val_acc:
        best_val_acc = val_acc
        torch.save(model.state_dict(), SAVE_PATH)
    print(f'Epoch [{epoch+1:02d}/{NUM_EPOCHS}] '
          f'Train Loss: {train_loss:.4f} Acc: {train_acc:.4f} | '
          f'Val Loss: {val_loss:.4f} Acc: {val_acc:.4f}'
          + (' <- best' if val_acc == best_val_acc else ''))

print(f'\nBest Val Accuracy: {best_val_acc:.4f}')
print(f'Saved: {SAVE_PATH}')
```

```
Epoch [01/30] Train Loss: 1.4770 Acc: 0.4286 | Val Loss: 1.5453 Acc: 0.3761 <
Epoch [02/30] Train Loss: 1.1969 Acc: 0.5321 | Val Loss: 1.1398 Acc: 0.5761 <
Epoch [03/30] Train Loss: 1.0652 Acc: 0.5709 | Val Loss: 1.3723 Acc: 0.5388
Epoch [04/30] Train Loss: 1.0095 Acc: 0.5843 | Val Loss: 0.9642 Acc: 0.6299 <
Epoch [05/30] Train Loss: 0.9916 Acc: 0.5982 | Val Loss: 1.1582 Acc: 0.5478
Epoch [06/30] Train Loss: 0.9264 Acc: 0.6405 | Val Loss: 0.9974 Acc: 0.6090
Epoch [07/30] Train Loss: 0.8633 Acc: 0.6599 | Val Loss: 1.1616 Acc: 0.5254
Epoch [08/30] Train Loss: 0.8030 Acc: 0.6773 | Val Loss: 0.8375 Acc: 0.6687 <
Epoch [09/30] Train Loss: 0.7537 Acc: 0.6917 | Val Loss: 1.3344 Acc: 0.5537
Epoch [10/30] Train Loss: 0.7439 Acc: 0.7101 | Val Loss: 0.8328 Acc: 0.6851 <
Epoch [11/30] Train Loss: 0.7549 Acc: 0.7190 | Val Loss: 1.6786 Acc: 0.5373
Epoch [12/30] Train Loss: 0.6684 Acc: 0.7350 | Val Loss: 0.8894 Acc: 0.6433
Epoch [13/30] Train Loss: 0.6099 Acc: 0.7534 | Val Loss: 0.8961 Acc: 0.6761
Epoch [14/30] Train Loss: 0.5353 Acc: 0.7946 | Val Loss: 0.8587 Acc: 0.6687
Epoch [15/30] Train Loss: 0.5032 Acc: 0.8031 | Val Loss: 0.8304 Acc: 0.7284 <
Epoch [16/30] Train Loss: 0.4325 Acc: 0.8359 | Val Loss: 0.7915 Acc: 0.7328 <
Epoch [17/30] Train Loss: 0.3635 Acc: 0.8642 | Val Loss: 0.7320 Acc: 0.7433 <
Epoch [18/30] Train Loss: 0.3304 Acc: 0.8777 | Val Loss: 0.7169 Acc: 0.7716 <
Epoch [19/30] Train Loss: 0.2956 Acc: 0.8961 | Val Loss: 0.7398 Acc: 0.7642
Epoch [20/30] Train Loss: 0.2026 Acc: 0.9304 | Val Loss: 0.8346 Acc: 0.7642
Epoch [21/30] Train Loss: 0.2095 Acc: 0.9219 | Val Loss: 0.6989 Acc: 0.7940 <
Epoch [22/30] Train Loss: 0.1560 Acc: 0.9493 | Val Loss: 0.7637 Acc: 0.7657
Epoch [23/30] Train Loss: 0.1339 Acc: 0.9577 | Val Loss: 0.8308 Acc: 0.7627
Epoch [24/30] Train Loss: 0.0914 Acc: 0.9717 | Val Loss: 0.7442 Acc: 0.8104 <
Epoch [25/30] Train Loss: 0.0880 Acc: 0.9702 | Val Loss: 0.8668 Acc: 0.7925
Epoch [26/30] Train Loss: 0.0643 Acc: 0.9836 | Val Loss: 0.8076 Acc: 0.7836
Epoch [27/30] Train Loss: 0.0477 Acc: 0.9881 | Val Loss: 0.8040 Acc: 0.7940
Epoch [28/30] Train Loss: 0.0506 Acc: 0.9861 | Val Loss: 0.8549 Acc: 0.7925
Epoch [29/30] Train Loss: 0.0489 Acc: 0.9876 | Val Loss: 0.8286 Acc: 0.8000
Epoch [30/30] Train Loss: 0.0455 Acc: 0.9901 | Val Loss: 0.8268 Acc: 0.7985

Best Val Accuracy: 0.8104
Saved: /content/drive/MyDrive/rice-convnext/02_GELU_StochDepth_BatchNorm/
convnext_tiny_02_gelu_stochdepth_batchnorm.pt
```

. Training Curves

In [11]:
```
epochs_range = range(1, NUM_EPOCHS + 1)
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].plot(epochs_range, history['train_loss'], label='Train Loss', col
axes[0].plot(epochs_range, history['val_loss'],   label='Val Loss',   col
axes[0].set_title('Loss', fontsize=13)
axes[0].set_xlabel('Epoch'); axes[0].set_ylabel('Loss')
axes[0].legend(); axes[0].grid(alpha=0.3)
axes[1].plot(epochs_range, history['train_acc'], label='Train Accuracy',
axes[1].plot(epochs_range, history['val_acc'],   label='Val Accuracy',
axes[1].set_title('Accuracy', fontsize=13)
axes[1].set_xlabel('Epoch'); axes[1].set_ylabel('Accuracy')
axes[1].legend(); axes[1].grid(alpha=0.3)
plt.suptitle(f'Training Curves — {EXP_NAME}', fontsize=11)
plt.tight_layout()
plt.savefig('training_curves.png', dpi=150)
plt.show()
```



. Evaluation on Test Set

In [12]:
```
model.load_state_dict(torch.load(SAVE_PATH, map_location=DEVICE))
model.eval()
all_preds, all_labels = [], []
with torch.no_grad():
    for images, labels in test_loader:
        images = images.to(DEVICE)
        outputs = model(images)
        _, predicted = outputs.max(1)
        all_preds.extend(predicted.cpu().numpy())
        all_labels.extend(labels.numpy())

report = classification_report(all_labels, all_preds, target_names=CLASS_
report_df = pd.DataFrame(report).transpose()
print('=== Classification Report ===')
print(report_df.round(4).to_string())
report_df.to_csv('classification_report.csv')

summary = {
    'Model'     : ['ConvNeXt-Tiny'],
    'Activation': ['GELU'],
    'Dropout'   : ['StochDepth'],
    'Norm'      : ['BatchNorm'],
    'Accuracy'  : [report['accuracy']],
    'Precision' : [report['weighted avg']['precision']],
```

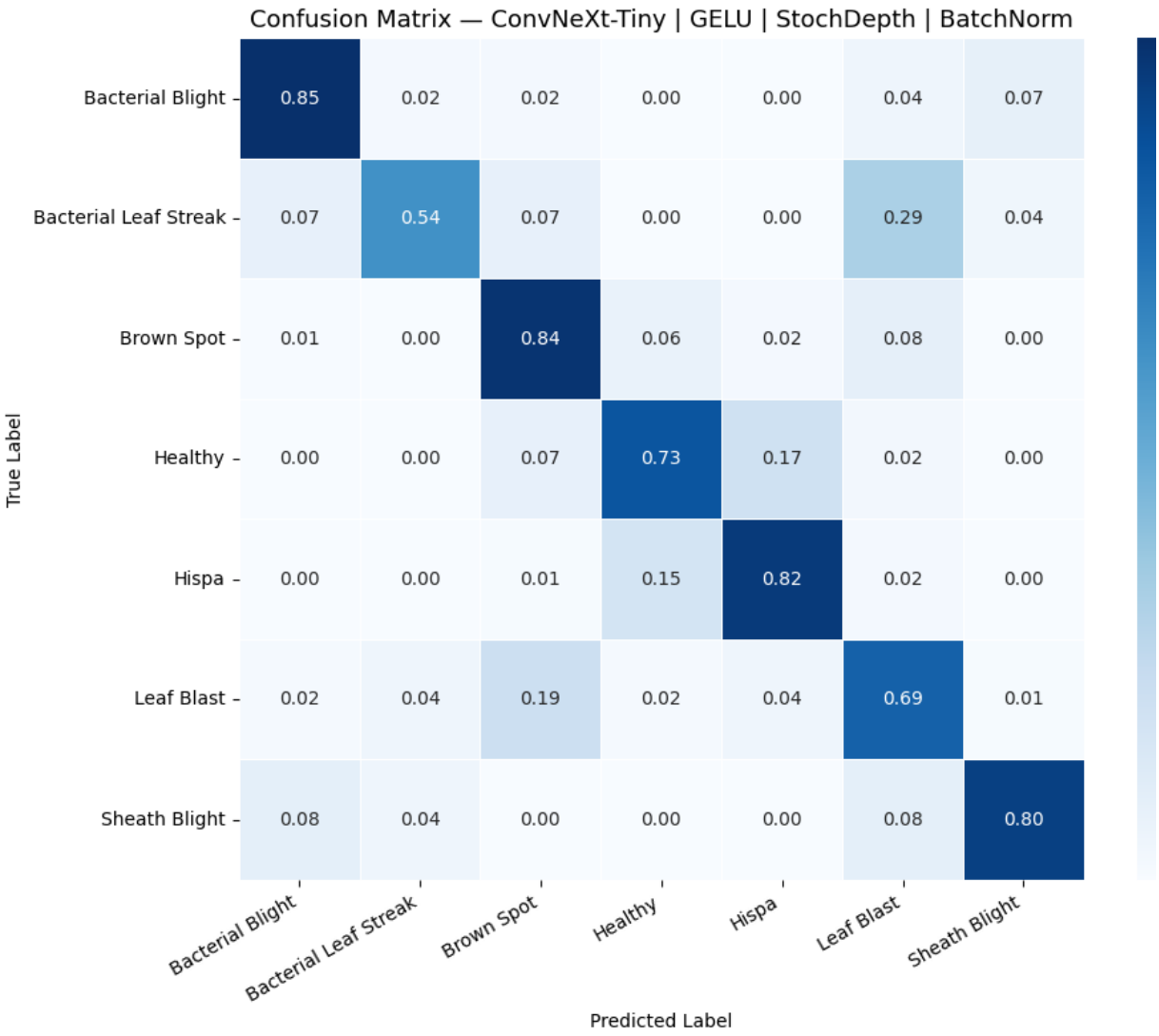
```
    'Recall'    : [report['weighted avg']['recall']],
    'F1-Score'  : [report['weighted avg']['f1-score']],
}
summary_df = pd.DataFrame(summary)
print('\n=== Summary ===')
print(summary_df.round(4).to_string(index=False))
summary_df.to_csv('summary.csv', index=False)
```

```
=== Classification Report ===
                      precision  recall  f1-score   support
Bacterial Blight         0.8679  0.8519    0.8598   54.0000
Bacterial Leaf Streak    0.6818  0.5357    0.6000   28.0000
Brown Spot               0.7809  0.8373    0.8081  166.0000
Healthy                  0.8067  0.7289    0.7658  166.0000
Hispa                    0.7087  0.8182    0.7595  110.0000
Leaf Blast               0.7328  0.6911    0.7113  123.0000
Sheath Blight            0.7692  0.8000    0.7843   25.0000
accuracy                 0.7679  0.7679    0.7679    0.7679
macro avg                0.7640  0.7519    0.7556  672.0000
weighted avg             0.7691  0.7679    0.7666  672.0000

=== Summary ===
        Model Activation    Dropout      Norm  Accuracy  Precision  Recall  F
Score
ConvNeXt-Tiny       GELU StochDepth BatchNorm    0.7679     0.7691  0.7679
0.7666
```

. Confusion Matrix

In [13]:

```
cm = confusion_matrix(all_labels, all_preds)
cm_norm = cm.astype('float') / cm.sum(axis=1)[:, np.newaxis]
plt.figure(figsize=(10, 8))
sns.heatmap(cm_norm, annot=True, fmt='.2f',
            xticklabels=CLASS_NAMES, yticklabels=CLASS_NAMES,
            cmap='Blues', linewidths=0.5)
plt.title(f'Confusion Matrix — ConvNeXt-Tiny | GELU | StochDepth | BatchN
plt.ylabel('True Label'); plt.xlabel('Predicted Label')
plt.xticks(rotation=30, ha='right')
plt.tight_layout()
plt.savefig('confusion_matrix.png', dpi=150)
plt.show()
```



. Save All to Drive

In [14]:
```
local_files = ['training_curves.png', 'confusion_matrix.png',
               'classification_report.csv', 'summary.csv', 'class_distrib
for fname in local_files:
    if os.path.exists(fname):
        shutil.copy(fname, f'{DRIVE_DIR}/{fname}')

all_outputs = [
    (f'{DRIVE_DIR}/convnext_tiny_02_gelu_stochdepth_batchnorm.pt',
    (f'{DRIVE_DIR}/training_curves.png',              'Training curves'),
    (f'{DRIVE_DIR}/confusion_matrix.png',             'Confusion matrix')
    (f'{DRIVE_DIR}/classification_report.csv',        'Per-class metrics'
    (f'{DRIVE_DIR}/summary.csv',                      'Summary table'),
]
print(f'=== Output Files: {DRIVE_DIR} ===')
for fpath, desc in all_outputs:
    exists = '✅' if os.path.exists(fpath) else '❌'
    print(f'{exists}  {os.path.basename(fpath):<45} {desc}')
```

```
=== Output Files: /content/drive/MyDrive/rice-convnext/02_GELU_StochDepth_Bat
===
✅  convnext_tiny_02_gelu_stochdepth_batchnorm.pt Model weights
✅  training_curves.png                           Training curves
✅  confusion_matrix.png                          Confusion matrix
✅  classification_report.csv                     Per-class metrics
✅  summary.csv                                   Summary table
```